Sum-Check Protocol

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Blockstream Research

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github.com/BlockstreamResearch/zk-book



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2 Multivariate Polynomials. Multilinear Extensions.

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Introduction

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Motivation

But what if we could work with much smaller degrees?

Sum-Check-based Protocols

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Sum-Check meaning

Instead of divisibility checks, we use the **Sum-Check**:

$$\sum_{b_1\in\{0,1\}} \sum_{b_2\in\{0,1\}} \cdots \sum_{b_v\in\{0,1\}} f(b_1, \ldots, b_v) = H$$

Summary in a Nutshell



Univariate World:

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Multivariate World:

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Succinct Arguments: Ligero (2022), Spartan (2020), Libra (2019), Hyrax(2017), GKR (2008), ...

Applications: zkGPT (2025), deep-prove (using *Ceno*) (2024), vSQL (2017), ...

Multivariate Polynomials. Multilinear Extensions.

Some technicalities

Definition (Monomial)

By **monomial** in v variables we call expression $\mu(\mathbf{X}) = X_1^{\alpha_1} \dots X_v^{\alpha_v}$ where $\alpha_1, \dots, \alpha_v$ are non-negative integers. The **degree** of a monomial is naturally defined as $\deg \mu \triangleq \alpha_1 + \dots + \alpha_v$.

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Function $f : \mathbb{F}^v \to \mathbb{F}$ is called an **v-variate polynomial**, which is denoted by $f \in \mathbb{F}[X_1, \dots, X_v]$, if $f(\mathbf{X})$ is a finite linear combination of v-variate monomials $\{\mu_1(\mathbf{X}), \dots, \mu_n(\mathbf{X})\}$. The **degree** of f is defined as $\deg f \triangleq \max_{i \in \{1,\dots,n\}} \deg \mu_i$.

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Example

$f(X_1, X_2, X_3) = X_1^3 + 3X_1^2X_2^2 + X_3^2 + X_3$ is a linear combination of 3-variate monomials $\{X_1^3, X_1^2X_2^2, X_3^2, X_3\}$, thus $f \in \mathbb{F}[X_1, X_2, X_3]$. It has degree $\deg f = 4$, corresponding to the second monomial $X_1^2X_2^2$.

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Similarly, $f(X_1, \ldots, X_v) = \prod_{i=1}^{v} X_i$ is a multilinear polynomial.

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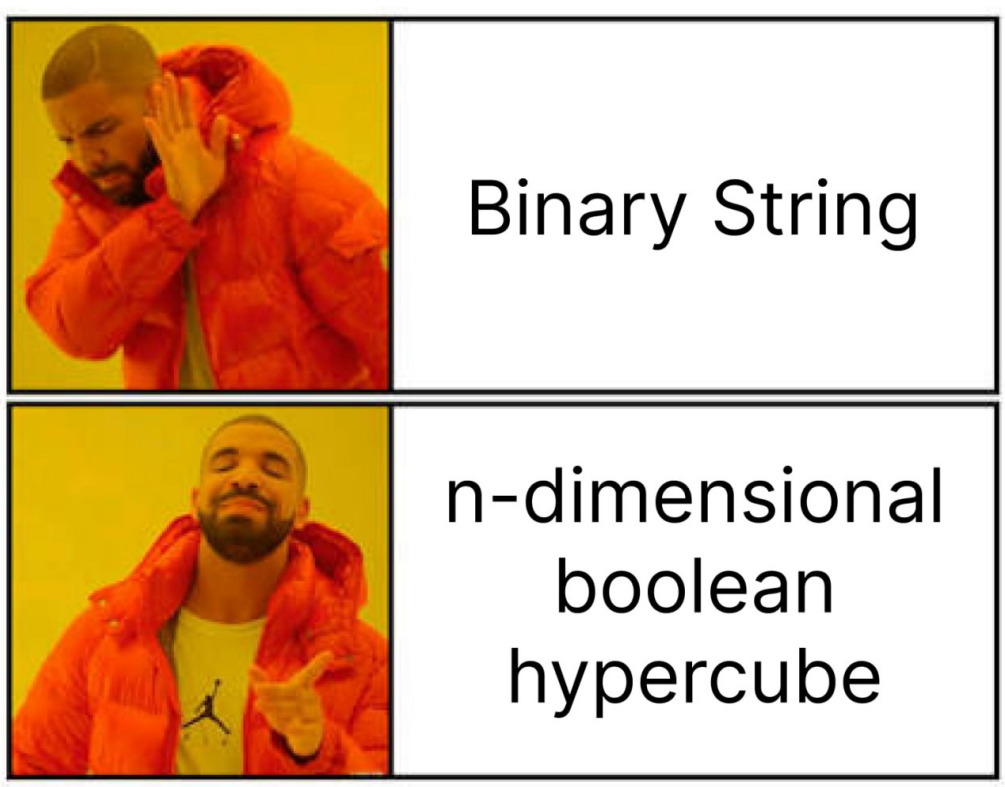


Figure: Cryptographers *love* overcomplicating things.

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Suppose we are given the values on the boolean hypercube $f : \{0,1\}^v \to \mathbb{F}$. We call $\widetilde{f} : \mathbb{F}^v \to \mathbb{F}$ an **extension** if $\widetilde{f}(\mathbf{b}) = f(\mathbf{b})$ for every $\mathbf{b} \in \{0,1\}^v$.

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Extend

	0	1	2	3	4
0	1	2	0	0	2
1	2	3	1	1	3
2	0	1	4	4	1
3	0	1	4	4	1
4	2	3	1	1	3

Extension $\widetilde{f} : \mathbb{F}_5^2 \to \mathbb{F}_5$

$\widetilde{f}(X_1, X_2) = X_1^2 + X_2^2 + 1$

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For the previous example $f(0,0) = 1, f(1,0) = f(0,1) = 2, f(1,1) = 3$ (over $\mathbb{F}_5$) the multilinear extension is given by $\widetilde{f}(X_1, X_2) = X_1 + X_2 + 1$.

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The question though is how many extensions $\widetilde{f}$ we can build.

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- However, if $\widetilde{f}$ is multilinear, it is **unique**.
- Additionally, is there an analogy to the *Lagrange Interpolation* for such case: how to build $\widetilde{f}$ practically?

Lagrange Interpolation of multilinear polynomials

Theorem (Lagrange Interpolation of Multilinear Polynomials)

Any function over the v-dimensional hypercube $f : \{0,1\}^v \to \mathbb{F}$ has a unique v-variate multilinear extension $\widetilde{f} \in \mathbb{F}[X_1, \ldots, X_v]$. It is defined using the **Lagrange interpolation of multilinear polynomials**:

$$\widetilde{f}(\mathrm{X}) = \sum_{\boldsymbol{b} \in \{0,1\}^v} f(\boldsymbol{b}) \cdot \mathrm{eq}(\mathrm{X}; \boldsymbol{b}), \quad \mathrm{eq}(\boldsymbol{b}'; \boldsymbol{b}) = \begin{cases} 1, & \boldsymbol{b}' = \boldsymbol{b} \\ 0, & \text{otherwise} \end{cases} \quad \text{for } \boldsymbol{b}' \in \{0,1\}^v,$$

where the set $\{\mathrm{eq}(\mathrm{X}; \boldsymbol{b})\}_{\boldsymbol{b} \in \{0,1\}^v}$ is referred to as **the set of multilinear Lagrange basis polynomials** over $\{0,1\}^v$.

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where the set $\{\mathrm{eq}(\mathrm{X}; \boldsymbol{b})\}_{\boldsymbol{b} \in \{0,1\}^v}$ is referred to as **the set of multilinear Lagrange basis polynomials** over $\{0,1\}^v$. Each basis polynomial (among 2^v) $\mathrm{eq}(\mathrm{X}; \boldsymbol{b})$ is defined as:

$$\mathrm{eq}(\mathrm{X}; \boldsymbol{b}) \triangleq \prod_{i=1}^{v} \{X_i b_i + (1 - X_i)(1 - b_i)\}.$$

Lagrange Interpolation: Example

Suppose we want to build the MLE for $f : \{0,1\}^2 \to \mathbb{F}_{11}$ given by:

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Step 1. Define multilinear Lagrange basis polynomials:

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Step 2. Find the appropriate linear combination:

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Fact: Generally, $\widetilde{f}(\boldsymbol{r})$ for $\boldsymbol{r} \leftarrow\!\$\ \mathbb{F}^{\nu}$ can be computed in $O(2^{\nu})$ time.

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Question

Where the hell should the Sum-Check be applied here?

Matrices Multilinear Extensions

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$$f_A(\boldsymbol{i}, \boldsymbol{j}) = A_{i,j}, \quad \text{where} \quad \boldsymbol{i} = (i_1, \ldots, i_{\log n}),\ \boldsymbol{j} = (j_1, \ldots, j_{\log n}).$$

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Its MLE is given by $\widetilde{f_A}(X, Y) = (1 - X)Y + 2X(1 - Y) = 2X + Y - 3XY$.

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Given functions $f_A, f_B, f_C : \{0,1\}^{\log n} \times \{0,1\}^{\log n} \to \mathbb{F}$, we build the corresponding MLEs $\widetilde{f_A}, \widetilde{f_B}, \widetilde{f_C} : \mathbb{F}^{\log n} \times \mathbb{F}^{\log n} \to \mathbb{F}$. Now what?

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Now, apply Sum-Check on $g(\mathsf{z}) = \widetilde{f_A}(\boldsymbol{r}_1, \mathsf{z}) \widetilde{f_B}(\mathsf{z}, \boldsymbol{r}_2)$ for $\boldsymbol{r}_1, \boldsymbol{r}_2 \leftarrow\$ \mathbb{F}^{\log n}$ to prove that it sums to $\widetilde{f_C}(\boldsymbol{r}_1, \boldsymbol{r}_2)$.

Idea of Spartan

This protocol might sound too abstract, but this idea of using matrix MLEs is used in **Spartan**! Recall that in QAP we check:

QAP Check: $$\sum_i z_i \ell_i(X) \cdot \sum_i z_i r_i(X) - \sum_i z_i o_i(X) = 0, \quad X \in \Omega$$

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4. Apply some dark magic to make above work.

Sum-Check Protocol

Sum-Check Protocol Goal

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Prover $\mathcal{P}$ wants to convince the verifier $\mathcal{V}$ that:

$$\sum_{b_1 \in \{0,1\}} \sum_{b_2 \in \{0,1\}} \cdots \sum_{b_v \in \{0,1\}} f(b_1, \ldots, b_v) = H$$

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Question

If f is multilinear, then what form does f_1 have?

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Assume prover sends $s_1(X_1)$. Verifier $\mathcal{V}$ needs to check:

1. $s_1(X_1)$ is indeed $f_1(X_1)$.
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For the second, apply the Schwartz-Zippel Lemma: pick $r_1 \leftarrow\!\!\$\ \mathbb{F}$ and check whether $s_1(r_1) = f_1(r_1)$. Computing $s_1(r_1)$ is trivial, but *how to compute* $f_1(r_1)$ *effectively*?

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Last Round. The verifier $\mathcal{V}$ picks a random $r_v \leftarrow\$ \mathbb{F}$ and verifies whether $s_v(r_v) = \mathcal{O}^f(r_1, \ldots, r_v)$ where $\mathcal{O}^f(\cdot)$ is an oracle access to the function f (e.g., commitment + opening).

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Lemma (Sum-Check Soundness Lemma)

Let $f \in \mathbb{F}[X_1, \ldots, X_v]$ be a multivariate polynomial of degree at most d in each variable, defined over the finite field $\mathbb{F}$. For any given $H \in \mathbb{F}$, let $\mathcal{L}$ be the language of all polynomials f (given as an oracle) such that

$$H = \sum_{b_1 \in \{0,1\}} \sum_{b_2 \in \{0,1\}} \cdots \sum_{b_v \in \{0,1\}} f(b_1, \ldots, b_v).$$

The sumcheck protocol is an IOP for $\mathcal{L}$ with the completeness error $\delta_C = 0$ and the soundness error $\delta_S \leq vd/\#\mathbb{F}$.

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Simply apply the **Fiat-Shamir heuristic**! At round j, the transcript is $\tau = (H, s_1, r_1, \ldots, s_{j-1}, r_{j-1}, s_j)$, thus the randomness can be sampled simply as $r_j \leftarrow \mathrm{H}(\tau)$ for a random oracle H.

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Now the coding time!

Thank you for your attention



research.blockstream.com

github.com/BlockstreamResearch/zk-book



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